

Elia Pattacini

eliapattacini0@gmail.com | +393343507488 | www.linkedin.com/in/elia-pattacini

Summary

Biomedical Engineer specialized in AI for time-series, image analysis and NLP with hands-on experience on large-scale time-series, predictive modeling, and production-oriented ML pipelines.

Skills

Programming: Python, MATLAB, RStudio, C++, SQL

Artificial intelligence Frameworks: Keras, Tensorflow, OpenCV, Pandas, Scikit-learn, Numpy

AI Architectures: Transformers, Convolutional Neural Networks (CNN), Neural Networks, LightGBM, Random Forest

Others: Microsoft Office Package, Git, PowerBI

Languages: Italian (Native), English (C1 - IELTS certification)

Experience

Data Scientist / Machine Learning Engineer, S.A.T.E. srlu - Venezia, Italy January 2025- present

- Worked with the collaboration of Eutelsat Group and ESA Connectivity and Secure Communications for the development of predictive diagnostics tools for spacecraft satellites anticipating the current detection of faulty systems by more than 3 months.
- Built end-to-end predictive diagnostics pipelines in Python (Pandas, NumPy, Scikit-learn, TensorFlow), from raw satellite telemetry to model validation and deployment-ready outputs.
- Performed large-scale time-series processing, feature engineering, and model evaluation using SQL/MATLAB and version control with Git.

Internship for Master thesis, Kaunas University of Technology(KTU)- Kaunas, Lithuania March 2023 – August 2023

- Development of a Deep Learning algorithm for the classification of ECG beat types between three categories (N,S,V) using a transformer architecture with a convolutional attention mechanism.
- Tested on different datasets, reaching a f1 score of 0.93 and a precision of 0.95.
- Worked with an international team for the implementation of artificially generated data in the training dataset

Projects

CNN for plant leaf classification

- Developed a CNN-based model for plant leaf classification achieving 98% accuracy on a 10,000+ image dataset while leading a team of four international students.

Development of AI Chat Bot Using Wizard of Wikipedia Dataset

- Fine-tuned a GPT-2 model for generating conversational responses using the "Wizard of Wikipedia" dataset.

Analysis of physiological signals for emotional recognition (Bachelor thesis)

- Study of three different physiological signals (ECG, RESP, SC) for emotional recognition on a healthy patient.

Education

Politecnico of Milan, Italy – MSc in Biomedical Engineering April 2024

(Master thesis: Custom Attention in Transformer Architectures for Multiclass ECG Beat Classification

University of Patras, Greece – Erasmus semester in Biomedical Engineering February 2023

Politecnico of Milan, Italy – BSc in Biomedical Engineering March 2021

(Bachelor thesis: Characterization of emotional states after auditory stimulus by means of physiological signals)

Notre Dame-Bishop Gibbons, Schenectady NY USA - High School Exchange Program June 2016

Publication

Pattacini E. et al., "Custom Attention in Transformer Architectures for Multiclass ECG Beat Classification", Springer, 2025. DOI: 10.1007/978-3-031-96538-8_9

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